NPRB2301: Estimation of shark body size from eye lenses (IN PROCESS)

January 12, 2026

Objective

The purpose of this analysis is to estimate body size of Pacific sleeper sharks and Pacific spiny dogfish from images of sequentially peeled eye lenses, based on the presumed linear relationship between eye lens diameter and total body length. The estimation of body size at the time of formation of each lens layer analyzed for delta C-14 and stable isotope composition is intended to facilitate estimation of growth rates and life history reconstruction of individual sharks.

Introduction

Vertebrate eye lenses grow continuously throughout life, adding sequential layers around an embryonic core over time. Other researchers have demonstrated linear relationships between the fish eye lens diameter and total length, including for the elasmobranchs porbeagle shark *Lamna nasus*, Atlantic spiny dogfish *Squalus acanthias*, thorny skate *Amblyraja radiata*, round ray *Rajella fyllae*, and leopard sharks *Triakis semifasciata* (Quaeck-Davies et al. 2018; Leifsdottir and Campana 2023; Kuntz 2025). In this subanalysis, we describe the relationship between whole lens diameter and the diameter of the hardened portion of the lens to the total length of Pacific sleeper sharks *Sominosus pacificus* and Pacific spiny dogfish *Squalus suckleyi* to estimate the total length at the time of formation of each peeled eye layer in order to link delta C-14 and stable isotope values to body size.

Methods

Imaging and measurements

Eye lenses were dissected from frozen eye globes and photographed against a 10 x 10 mm grid background to spatially calibrate images for measurements (Figure 1). For most individuals, a photograph was taken of the whole, intact lens and of the lens following each sequential peel to a small central nucleus. Eye lenses differed in appearance. While the majority of whole lenses were spherical, a few were oblong in shape; many appeared firmer and more opaque, whereas other lenses were surrounded with a clear jelly layer (Figure 2). Measurements were generally taken through the equator at the widest point of the solid lens, unless the capsule was clearly beginning to delaminate from the rest of the lens. In those cases, the lens was measured where it appeared most spherical (Figure 3). Some lenses were too irregular to measure. The condition of the lens was noted when taking measurements. All measurements were made using ImagePro v.10-11 (Media Cybernetics) and converted from pixels to mm in R statistical computing software.

Comparison of capsule diameter to the diamter of the first hard layer

The outermost layer of the whole lens is termed the "capsule". Just medial of the capsule is a layer of jelly-like cuboidal epithelial cells, followed by hardened layers composed of crystalline proteins (Kuntz et al. 2025, Quack-Davies et al. 2018). Studies have suggested that the ratio of the whole lens diameter (LD_w) and the first hardened layer (LD_h) is consistent across the life of the fish, regardless of species (Leiffsdottir and Campana 2023). A regression of the LD_h against the LD_w was conducted by species, and a t-test was performed to test if the slopes of the lines were significantly different from 1. A slope of 1 would be expected if the LD_w and LD_h increased equally as the eye grows. Similarly, we also examined the relationships of the LD_h:LD_w to total body length at the time of capture (TL_c) and the LD_h:LD_w to LD_w, but tested if those slopes are significantly different from zero. In these cases a slope of zero would be expected if the ratio of LD_h:LD_w was conserved as the eye or fish grows.

Relationship of LD_w and LD_h to TL

The relationships between LD_w, LD_h, and TLc was described using separate linear models for each species and layer diameter type. The equation followed the form:

$$LD = a_i, j + TL_c * b_i, j$$

Where LD is either LD_w or LD_h, i is the species and j is either w or h for the LD_w or LD_h regression. The slope and intercept parameters are b and a, respectively.

Images of samples with observations outside the 99% confidence limits of the fitted LD_w or LD_h ~ TLc relationships were re-examined for measurement errors. The R package 'olsrr' was used to examine residual patterns, check for highly influential observations, and run diagnostics.

Estimating fish length at the time of layer formation

The above equation can be rewritten to estimate the total length at the time of layer formation (TL_t) following Kuntz (2025):

$$TL_t = (LD_t - a_i, j)/b_i, j$$

where LD_t is the diameter of the lens following each sequential peel and the $a_{i,j}$ and $b_{i,j}$ parameters are from Table 2.

Results

Comparison of capsule diameter to the diamter of the first hard layer

Paired LD_w and LD_h were available from 36 Pacific sleeper sharks and 7 Pacific spiny dogfish with total length at time of capture. The slope of the regression line between LD_h and LD_w was significantly different from 1 (alpha = 0.05) for Pacific Sleeper Shark, but not Pacific Spiny Dogfish (Table 1, Model = "LDh vs LDw", Figure 4). The slopes of the regression lines of the ratio of LD_h:LD_w relative to the TL_c and relative to the LD_w were significantly different from zero for Pacific sleeper sharks, but not for Pacific spiny dogfish (Table 1, Model = "ratio vs TLc" and "ratio vs LDw", respectively, Figure 5).

Relationship of LD_w and LD_h to TL

Paired LDw and TLc data were available from 54 Pacific sleeper sharks and 20 spiny dogfish, while paired LDh and TLc data were was available from 52 Pacific sleeper sharks and 52 spiny dogfish. NEED TO DO THE influential evaluations still

Estimatig fish length at the time of layer formation

Tables

Table 1: Linear regression slope statistics, with 95% confidence intervals, standard error, Rsquared, degrees of freedom, t-test p-value at alpha = 0.05, the model and the hypothesis that was tested

Species	Slope	Lower	Upper	StErr	Rsq	DF	Pval	Model	Hypothesis
SD	0.5913	-0.0337	1.2163	0.2432	0.5419	5	0.1536	LDh vs LDw	mu = 1
PSS	0.6559	0.5544	0.7574	0.0499	0.8353	34	0.0000	LDh vs LDw	mu = 1
SD	-0.0005	-0.0474	0.0464	0.0183	0.0001	5	0.9799	ratio vs LDw	mu = 0
PSS	0.0159	0.0070	0.0248	0.0044	0.2806	34	0.0009	ratio vs LDw	mu = 0
SD	-0.0023	-0.0054	0.0008	0.0012	0.4232	5	0.1136	ratio vs TL	mu = 0
PSS	0.0005	0.0003	0.0007	0.0001	0.3448	34	0.0002	ratio vs TL	mu = 0

Table 2: Linear regression statistics of the whole lens diameter (LD_w) and the first hardened layer (LD_h) to total length at capture (TL_c) for each species (i), with 95% confidence intervals, standard error, Rsquared, degrees of freedom, t-test p-value at alpha = 0.05, and the model (j).

Species	Parameter	Value	Lower	Upper	StErr	Rsq	DF	Pval	Model
SD	a	-34.1385	-46.3812	-21.8958	5.8273	0.9335	18	0.0000	LDw vs TL
SD	b	10.1698	8.8260	11.5136	0.6396	0.9335	18	0.0000	LDw vs TL
PSS	a	-125.6817	-172.6547	-78.7088	23.4087	0.7990	52	0.0000	LDw vs TL
PSS	b	31.0183	26.6888	35.3478	2.1576	0.7990	52	0.0000	LDw vs TL
SD	a	-21.4138	-32.0659	-10.7617	5.1066	0.9294	20	0.0004	LDh vs TL
SD	b	14.6961	12.8073	16.5848	0.9055	0.9294	20	0.0000	LDh vs TL
PSS	a	-0.8731	-30.8355	29.0894	14.9174	0.7723	50	0.9536	LDh vs TL
PSS	b	43.1987	36.5350	49.8625	3.3177	0.7723	50	0.0000	LDh vs TL

Figures

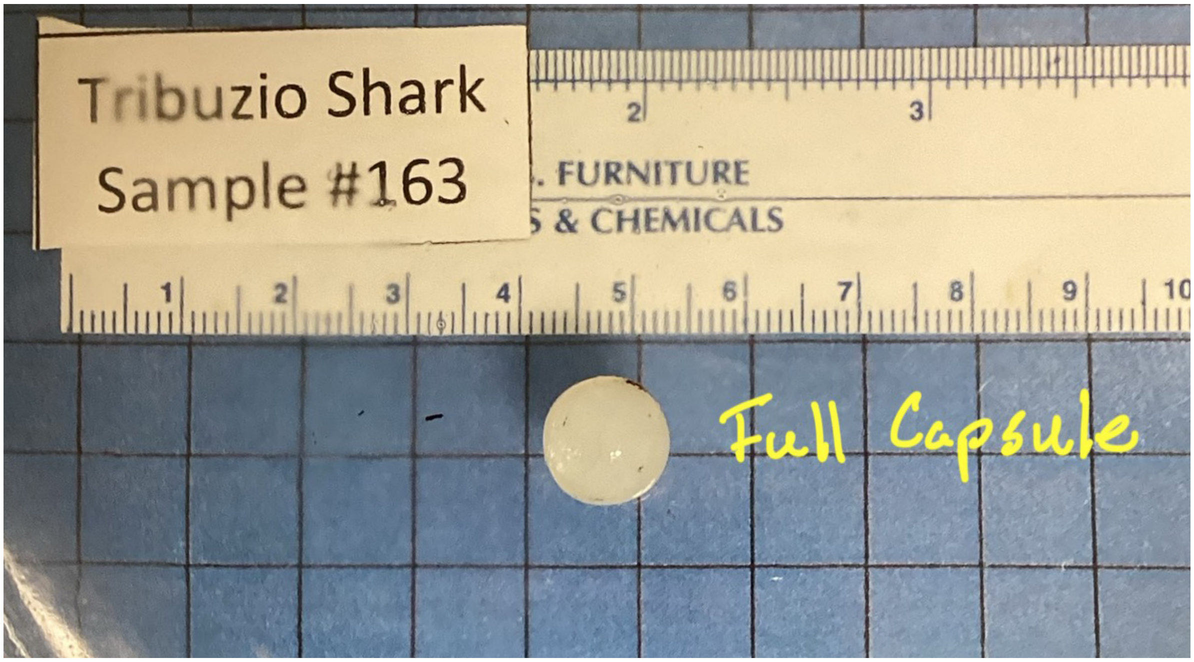

Figure 1: Image showing a whole, intact eye lens against gridded 10 x 10 mm background.

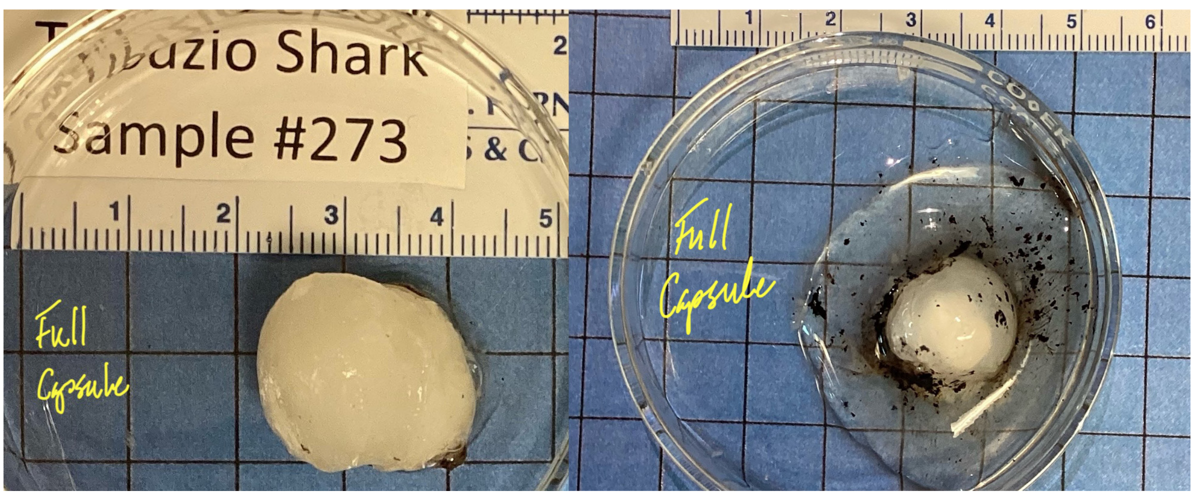

Figure 2: Eye lenses from 2 sharks of about the same length (left = 371 cm TL, right = 403 cm TL). Note the oblong shape of the lens on left and jelly layer surrounding the lens on right.

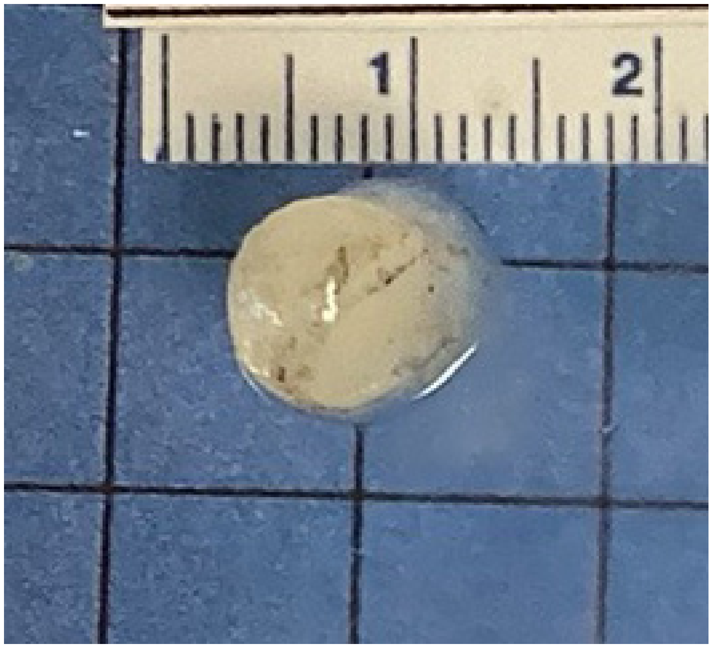

Figure 3: Outer capsule starting to separate from lens. Lens was measured where it is most spherical.

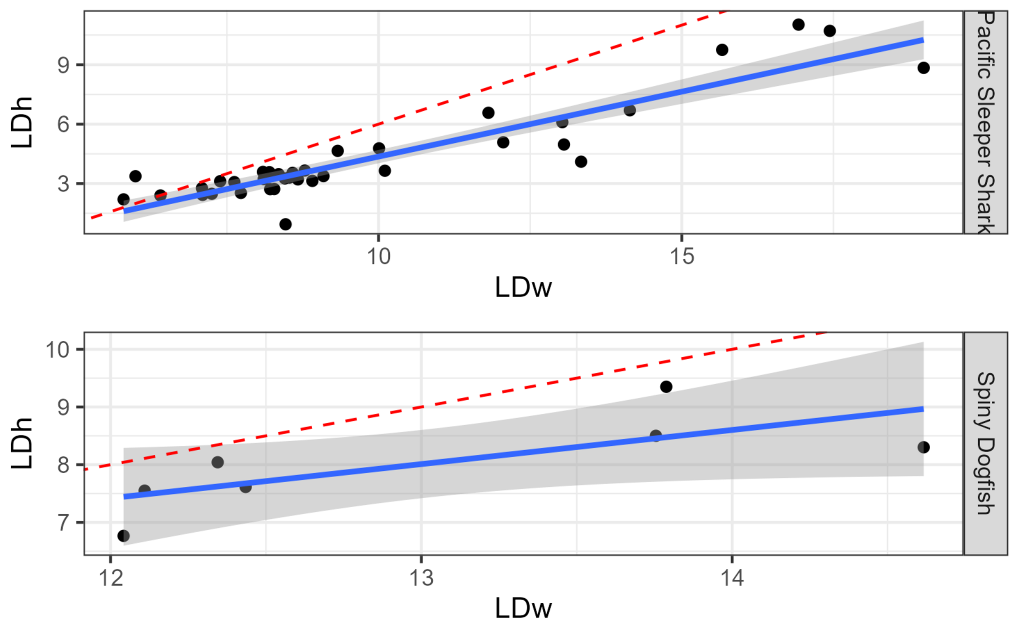

Figure 4: Linear regressions of the diameter of the whole eye (LDw) and the first hardened layer (LDh). Shaded region represents the 95% confidence interval. The red dashed line is the expected 1:1 replacement line, null hypothesis mu = 1.

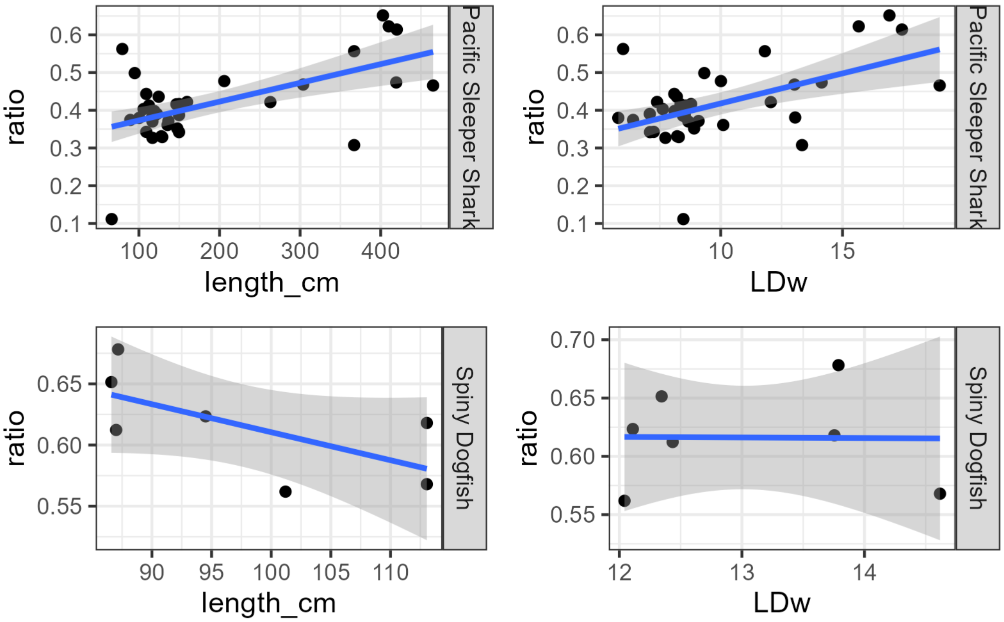

Figure 5: Linear regressions of the ratio of the LDh to LDw relative to the total length at capture and to the LDw. Shaded region represents the 95% confidence interval.

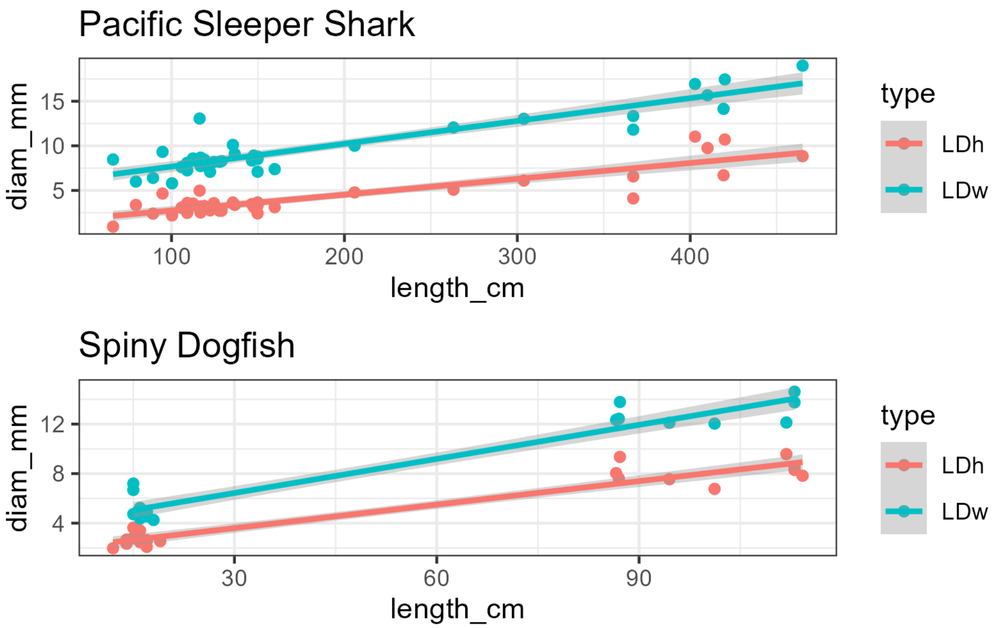

Figure 6: Linear regressions of the LDh and the LDw to the total length at capture. Shaded region represents the 95% confidence interval.